

Chapter 1 Exercises

SOLUTIONS TO RICK DURRETT'S
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 1 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter1_Knowledge`. The cited knowledge IDs refer to entries in `chapter1_knowledge_pieces.jsonl`.

Exercise 1.1.1

Question. Let $\Omega = \mathbb{R}$, let $\mathcal{F}$ be the collection of all subsets A such that A or A^c is countable, and define

$$P(A) = \begin{cases} 0, & A \text{ is countable,} \\ 1, & A^c \text{ is countable.} \end{cases}$$

Show that $(\Omega, \mathcal{F}, P)$ is a probability space.

Knowledge used. `durrett_ch1_probability_space`; `durrett_ch1_sigma_field`; `durrett_ch1_measure_continuity`.

Solution. First we show that $\mathcal{F}$ is a σ -field. Since $\emptyset$ is countable, $\emptyset \in \mathcal{F}$ and $\Omega \in \mathcal{F}$. If $A \in \mathcal{F}$, then either A is countable or A^c is countable, so the same alternative holds for A^c ; hence $A^c \in \mathcal{F}$.

Now let $A_1, A_2, \dots \in \mathcal{F}$. If every A_n is countable, then $\bigcup_n A_n$ is countable, so $\bigcup_n A_n \in \mathcal{F}$. If at least one A_k is uncountable, then A_k^c is countable. Since

$$\left(\bigcup_{n=1}^{\infty} A_n \right)^c = \bigcap_{n=1}^{\infty} A_n^c \subseteq A_k^c,$$

the complement of the union is countable. Therefore $\bigcup_n A_n \in \mathcal{F}$. Thus $\mathcal{F}$ is a σ -field.

The definition of P is consistent: no set can be both countable and co-countable in $\mathbb{R}$, because $\mathbb{R}$ is uncountable. Also $P(\Omega) = 1$. It remains to prove countable additivity. Let $A_1, A_2, \dots$ be pairwise disjoint members of $\mathcal{F}$.

If all A_n are countable, then $\bigcup_n A_n$ is countable and

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = 0 = \sum_{n=1}^{\infty} P(A_n).$$

If some A_k is co-countable, then no other disjoint A_n can be co-countable. Indeed, for $n \neq k$ we have $A_n \subseteq A_k^c$, so A_n is countable. Hence

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = 1 = P(A_k) + \sum_{n \neq k} P(A_n).$$

Therefore P is countably additive and $(\Omega, \mathcal{F}, P)$ is a probability space.

Exercise 1.1.2

Question. Recall the definition of $\mathcal{S}_d$ from Example 1.1.5:

$$\mathcal{S}_d = \{\emptyset\} \cup \{(a_1, b_1] \times \cdots \times (a_d, b_d] : -\infty \leq a_i < b_i \leq \infty\}.$$

Show that $\sigma(\mathcal{S}_d) = \mathcal{B}(\mathbb{R}^d)$, the Borel subsets of $\mathbb{R}^d$.

Knowledge used. `durrett_ch1_sigma_field`; `durrett_ch1_semi-algebra-extension`; `durrett_ch1-product-distribution-function`.

Solution. Every set in $\mathcal{S}_d$ is a product of half-open intervals, each of which is Borel in $\mathbb{R}$. Hence every element of $\mathcal{S}_d$ is Borel in $\mathbb{R}^d$, and

$$\sigma(\mathcal{S}_d) \subseteq \mathcal{B}(\mathbb{R}^d).$$

For the reverse inclusion, it is enough to show that every open subset of $\mathbb{R}^d$ lies in $\sigma(\mathcal{S}_d)$, because the Borel σ -field is generated by the open sets. First observe that if $q_i < r_i$ are rational numbers, then the open rational rectangle

$$Q = (q_1, r_1) \times \cdots \times (q_d, r_d)$$

belongs to $\sigma(\mathcal{S}_d)$. Indeed,

$$Q = \bigcup_{m=1}^{\infty} (q_1, r_1 - 1/m] \times \cdots \times (q_d, r_d - 1/m],$$

where terms with $r_i - 1/m \leq q_i$ are simply omitted. Each displayed half-open rectangle belongs to $\mathcal{S}_d$, so $Q \in \sigma(\mathcal{S}_d)$.

Now let $G \subseteq \mathbb{R}^d$ be open. Since $\mathbb{R}^d$ has a countable base of open rectangles with rational endpoints, G is a countable union of such rational rectangles. Each rational rectangle is in $\sigma(\mathcal{S}_d)$, so $G \in \sigma(\mathcal{S}_d)$. Thus $\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\mathcal{S}_d)$. Combining both inclusions gives $\sigma(\mathcal{S}_d) = \mathcal{B}(\mathbb{R}^d)$.

Exercise 1.1.3

Question. A σ -field $\mathcal{F}$ is said to be countably generated if there is a countable collection $\mathcal{C} \subseteq \mathcal{F}$ such that $\sigma(\mathcal{C}) = \mathcal{F}$. Show that $\mathcal{B}(\mathbb{R}^d)$ is countably generated.

Knowledge used. `durrett_ch1_sigma_field`; `durrett_ch1_semi-algebra-extension`; `durrett_ch1-product-distribution-function`.

Solution. Let $\mathcal{C}$ be the collection of all open balls in $\mathbb{R}^d$ whose centers have rational coordinates and whose radii are positive rational numbers:

$$\mathcal{C} = \{B(q, r) : q \in \mathbb{Q}^d, r \in \mathbb{Q}, r > 0\}.$$

This collection is countable because $\mathbb{Q}^d \times \mathbb{Q}_+$ is countable.

We claim that $\mathcal{C}$ is a base for the usual topology on $\mathbb{R}^d$. Let $G \subseteq \mathbb{R}^d$ be open and let $x \in G$. Choose $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq G$. Pick a rational point $q \in \mathbb{Q}^d$ close enough to x and a rational radius $r > 0$ small enough so that

$$x \in B(q, r) \subseteq B(x, \varepsilon) \subseteq G.$$

Therefore every open set is a union of sets from $\mathcal{C}$. Since $\mathcal{C}$ is countable, every open set is in fact a countable union of members of $\mathcal{C}$.

Thus all open sets belong to $\sigma(\mathcal{C})$, so $\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\mathcal{C})$. Conversely, every member of $\mathcal{C}$ is open and hence Borel, so $\sigma(\mathcal{C}) \subseteq \mathcal{B}(\mathbb{R}^d)$. Therefore

$$\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R}^d),$$

and $\mathcal{B}(\mathbb{R}^d)$ is countably generated.

Exercise 1.1.4

Question.

- (i) Show that if $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$ are σ -algebras, then $\bigcup_i \mathcal{F}_i$ is an algebra.
- (ii) Give an example to show that $\bigcup_i \mathcal{F}_i$ need not be a σ -algebra.

Knowledge used. [durrett_ch1_sigma_field](#).

Solution. Let

$$\mathcal{F}_\infty = \bigcup_{i=1}^{\infty} \mathcal{F}_i.$$

First, $\Omega \in \mathcal{F}_1 \subseteq \mathcal{F}_\infty$. If $A \in \mathcal{F}_\infty$, then $A \in \mathcal{F}_i$ for some i . Since $\mathcal{F}_i$ is a σ -algebra, $A^c \in \mathcal{F}_i \subseteq \mathcal{F}_\infty$.

Now let $A, B \in \mathcal{F}_\infty$. Then $A \in \mathcal{F}_i$ and $B \in \mathcal{F}_j$ for some i, j . If $k = \max(i, j)$, then $A, B \in \mathcal{F}_k$ because the sequence is increasing. Hence $A \cup B \in \mathcal{F}_k \subseteq \mathcal{F}_\infty$. Thus $\mathcal{F}_\infty$ is closed under complements and finite unions, so it is an algebra.

For a counterexample to countable closure, take $\Omega = \mathbb{N}$ and let

$$\mathcal{F}_n = \sigma(\{1\}, \{2\}, \dots, \{n\}).$$

Equivalently, $\mathcal{F}_n$ consists of all sets whose behavior on $\{1, \dots, n\}$ is arbitrary but whose tail $\{n+1, n+2, \dots\}$ is either entirely included or entirely excluded. Then

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$$

are σ -algebras. Their union contains every finite subset of $\mathbb{N}$, because any finite set is contained in $\{1, \dots, n\}$ for large enough n . Hence each singleton $\{2k\}$ belongs to $\bigcup_n \mathcal{F}_n$. But

$$E = \{2, 4, 6, \dots\} = \bigcup_{k=1}^{\infty} \{2k\}$$

is neither finite nor cofinite, so it does not belong to any $\mathcal{F}_n$. Therefore $\bigcup_n \mathcal{F}_n$ is not closed under countable unions and is not a σ -algebra.

Exercise 1.1.5

Question. A set $A \subseteq \{1, 2, \dots\}$ is said to have asymptotic density θ if

$$\lim_{n \rightarrow \infty} \frac{|A \cap \{1, 2, \dots, n\}|}{n} = \theta.$$

Let $\mathcal{A}$ be the collection of sets for which the asymptotic density exists. Is $\mathcal{A}$ a σ -algebra? Is $\mathcal{A}$ an algebra?

Knowledge used. `durrett_ch1_sigma_field`; `durrett_ch1_measure_continuity`.

Solution. The collection $\mathcal{A}$ is not an algebra, and therefore it is not a σ -algebra either. We construct two sets with asymptotic density but whose intersection does not have asymptotic density.

Let $N_0 = 0$ and choose a rapidly increasing sequence, for example $N_k = 2^{2^k}$. Define blocks

$$I_k = \{N_{k-1} + 1, \dots, N_k\}, \quad k \geq 1.$$

Let

$$E = \{2, 4, 6, \dots\}$$

be the even integers, so E has asymptotic density $1/2$.

Now define a set B block by block:

$$B \cap I_k = \begin{cases} E \cap I_k, & k \text{ odd,} \\ E^c \cap I_k, & k \text{ even.} \end{cases}$$

In every block, B contains exactly one parity class, so up to a bounded rounding error it contains half of each block. Consequently

$$\frac{|B \cap \{1, \dots, n\}|}{n} \rightarrow \frac{1}{2},$$

and hence $B \in \mathcal{A}$.

Consider now $E \cap B$. On odd blocks, $E \cap B$ contains essentially all even numbers in the block; on even blocks, it contains none of the even numbers in the block. Because $N_k/N_{k-1} \rightarrow \infty$, the most recent block dominates all previous blocks. Therefore, along the end of odd blocks,

$$\frac{|(E \cap B) \cap \{1, \dots, N_k\}|}{N_k} \rightarrow \frac{1}{2},$$

whereas along the end of even blocks,

$$\frac{|(E \cap B) \cap \{1, \dots, N_k\}|}{N_k} \rightarrow 0.$$

The density of $E \cap B$ does not exist.

Thus $E \in \mathcal{A}$ and $B \in \mathcal{A}$, but $E \cap B \notin \mathcal{A}$. Hence $\mathcal{A}$ is not closed under finite intersections, so it is not an algebra. Since every σ -algebra is an algebra, $\mathcal{A}$ is not a σ -algebra.

Exercise 1.2.1

Question. Let X and Y be random variables on $(\Omega, \mathcal{F}, P)$ and let $A \in \mathcal{F}$. Define $Z = X$ on A and $Z = Y$ on A^c . Show that Z is a random variable.

Knowledge used. `durrett_ch1_measurable_map`; `durrett_ch1_measurability_generator`; `durrett_ch1_sigma_field`.

Solution. It is enough to check that $\{Z \leq x\} \in \mathcal{F}$ for each $x \in \mathbb{R}$. For any real x ,

$$\{Z \leq x\} = (A \cap \{X \leq x\}) \cup (A^c \cap \{Y \leq x\}).$$

Since $A \in \mathcal{F}$, $A^c \in \mathcal{F}$, and X, Y are random variables, both $\{X \leq x\}$ and $\{Y \leq x\}$ are in $\mathcal{F}$. The right-hand side is formed using finite intersections and a finite union, so it belongs to $\mathcal{F}$. Therefore Z is measurable and hence is a random variable.

Exercise 1.2.2

Question. Let χ be standard normal. Use Durrett's normal-tail estimate to find upper and lower bounds for $P(\chi \geq 4)$.

Knowledge used. [durrett_ch1_normal_tail_bound](#); [durrett_ch1_distribution_function](#).

Solution. For a standard normal random variable,

$$P(\chi \geq 4) = \frac{1}{\sqrt{2\pi}} \int_4^\infty e^{-y^2/2} dy.$$

Durrett's Theorem 1.2.6 says that, for $x > 0$,

$$(x^{-1} - x^{-3})e^{-x^2/2} \leq \int_x^\infty e^{-y^2/2} dy \leq x^{-1}e^{-x^2/2}.$$

With $x = 4$, this gives

$$\frac{1}{\sqrt{2\pi}} \left(\frac{1}{4} - \frac{1}{64} \right) e^{-8} \leq P(\chi \geq 4) \leq \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{4} e^{-8}.$$

Equivalently,

$$\boxed{\frac{15}{64\sqrt{2\pi}} e^{-8} \leq P(\chi \geq 4) \leq \frac{1}{4\sqrt{2\pi}} e^{-8}}$$

which is approximately

$$3.135 \times 10^{-5} \leq P(\chi \geq 4) \leq 3.346 \times 10^{-5}.$$

Exercise 1.2.3

Question. Show that a distribution function has at most countably many discontinuities.

Knowledge used. [durrett_ch1_distribution_function](#); [durrett_ch1_atoms_jumps](#); [durrett_ch1_measure_continuity](#).

Solution. Let F be a distribution function. Since F is nondecreasing, the left limit

$$F(x-) = \lim_{y \uparrow x} F(y)$$

exists for every x . A discontinuity of a distribution function is a jump, so x is a discontinuity exactly when

$$\Delta F(x) := F(x) - F(x-) > 0.$$

For integers $m, n \geq 1$, define

$$D_{m,n} = \{x \in [-n, n] : \Delta F(x) > 1/m\}.$$

We claim $D_{m,n}$ is finite. If $x_1 < \dots < x_k$ are distinct points in $D_{m,n}$, then the jumps at these points are disjoint contributions to the total increase of F on an interval containing $[-n, n]$. Hence

$$\sum_{j=1}^k \Delta F(x_j) \leq 1.$$

But each jump is greater than $1/m$, so $k < m$. Thus $D_{m,n}$ is finite.

Every discontinuity has a positive jump, so it lies in $D_{m,n}$ for some m, n . Therefore the set of discontinuities is contained in

$$\bigcup_{m=1}^{\infty} \bigcup_{n=1}^{\infty} D_{m,n},$$

a countable union of finite sets. Hence a distribution function has at most countably many discontinuities.

Exercise 1.2.4

Question. Suppose $F(x) = P(X \leq x)$ is continuous. Show that $Y = F(X)$ is uniform on $(0, 1)$, i.e. $P(Y \leq y) = y$ for $0 \leq y \leq 1$.

Knowledge used. `durrett_ch1_probability_integral_transform`;
`durrett_ch1_distribution_function`; `durrett_ch1_cdf_to_law`.

Solution. Fix $y \in [0, 1]$. Define

$$x_y = \sup\{x \in \mathbb{R} : F(x) \leq y\}.$$

Using the continuity of F and the limits $F(-\infty) = 0$, $F(\infty) = 1$, we have $F(x_y) = y$, with the obvious interpretation at $y = 0$ and $y = 1$.

Because F is nondecreasing,

$$\{F(X) \leq y\} = \{X \leq x_y\}$$

up to possible endpoints in flat parts of F ; continuity ensures those endpoints do not change the probability beyond the value $F(x_y)$. More formally, the set $\{x : F(x) \leq y\}$ is a left ray ending at x_y , possibly with a flat interval at level y , and its probability is $F(x_y) = y$. Therefore

$$P(Y \leq y) = P(F(X) \leq y) = P(X \leq x_y) = F(x_y) = y.$$

Thus Y has the uniform distribution on $(0, 1)$.

Exercise 1.2.5

Question. Suppose X has continuous density f and is supported on $[\alpha, \beta]$. Let g be strictly increasing and differentiable on (α, β) . Find the density of $g(X)$.

Knowledge used. `durrett_ch1_change_of_variables`; `durrett_ch1_distribution_function`;
`durrett_ch1_cdf_to_law`.

Solution. Let $Y = g(X)$. Since g is strictly increasing, it has an inverse on $(g(\alpha), g(\beta))$. For $y \in (g(\alpha), g(\beta))$,

$$P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = \int_{\alpha}^{g^{-1}(y)} f(x) dx.$$

Differentiating with respect to y gives

$$f_Y(y) = f(g^{-1}(y)) \frac{d}{dy} g^{-1}(y) = \frac{f(g^{-1}(y))}{g'(g^{-1}(y))}, \quad g(\alpha) < y < g(\beta),$$

where the displayed formula is understood at points where $g'(g^{-1}(y)) > 0$. Outside the interval $(g(\alpha), g(\beta))$, the density is 0.

In particular, if $g(x) = ax + b$ with $a > 0$, then $g^{-1}(y) = (y - b)/a$ and

$$f_Y(y) = \frac{1}{a} f\left(\frac{y - b}{a}\right).$$

Exercise 1.2.6

Question. Suppose X is normally distributed. Use the previous exercise to find the density of $\exp(X)$, the lognormal distribution.

Knowledge used. `durrett_ch1_change_of_variables`;
`durrett_ch1_common_distributions_expectations`; `durrett_ch1_distribution_function`.

Solution. Let $X \sim N(\mu, \sigma^2)$ with $\sigma > 0$, and set

$$Y = e^X.$$

Here $g(x) = e^x$, so $g^{-1}(y) = \log y$ and $g'(g^{-1}(y)) = e^{\log y} = y$. The density of X is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

Therefore, for $y > 0$,

$$f_Y(y) = \frac{f_X(\log y)}{y} = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log y - \mu)^2}{2\sigma^2}\right].$$

For $y \leq 0$, $f_Y(y) = 0$. Thus

$$f_Y(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log y - \mu)^2}{2\sigma^2}\right] \mathbf{1}_{(0,\infty)}(y)$$

is the lognormal density.

Exercise 1.2.7

Question.

- (i) If X has density f , compute the distribution function and density of X^2 .
- (ii) Specialize to X standard normal to get the chi-square density with one degree of freedom.

Knowledge used. `durrett_ch1_change_of_variables`; `durrett_ch1_distribution_function`;
`durrett_ch1_common_distributions_expectations`.

Solution. Let $Y = X^2$. For $y < 0$, clearly $P(Y \leq y) = 0$. For $y \geq 0$,

$$F_Y(y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} f(x) dx.$$

Differentiating for $y > 0$ gives

$$f_Y(y) = f(\sqrt{y}) \frac{1}{2\sqrt{y}} + f(-\sqrt{y}) \frac{1}{2\sqrt{y}} = \frac{f(\sqrt{y}) + f(-\sqrt{y})}{2\sqrt{y}}, \quad y > 0.$$

Thus

$$f_Y(y) = \frac{f(\sqrt{y}) + f(-\sqrt{y})}{2\sqrt{y}} \mathbf{1}_{(0,\infty)}(y).$$

If X is standard normal, then

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

so $f(\sqrt{y}) = f(-\sqrt{y}) = (2\pi)^{-1/2} e^{-y/2}$. Hence, for $y > 0$,

$$f_Y(y) = \frac{2(2\pi)^{-1/2} e^{-y/2}}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}.$$

Therefore X^2 has density

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-y/2} \mathbf{1}_{(0,\infty)}(y)$$

which is the chi-square distribution with one degree of freedom.

Exercise 1.3.1

Question. Show that if $\mathcal{A}$ generates $\mathcal{S}$, then

$$X^{-1}(\mathcal{A}) \equiv \{\{X \in A\} : A \in \mathcal{A}\}$$

generates

$$\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}.$$

Knowledge used. `durrett_ch1_measurability_generator`; `durrett_ch1_sigma_field`; `durrett_ch1_measurable_map`.

Solution. Let

$$\mathcal{G} = \sigma(X^{-1}(\mathcal{A})).$$

Since $\mathcal{A} \subseteq \mathcal{S}$, we immediately have $X^{-1}(\mathcal{A}) \subseteq \sigma(X)$, so

$$\mathcal{G} \subseteq \sigma(X).$$

For the reverse inclusion, define

$$\mathcal{D} = \{B \in \mathcal{S} : \{X \in B\} \in \mathcal{G}\}.$$

The class $\mathcal{D}$ is a σ -field: inverse images commute with complements and countable unions. Also $\mathcal{A} \subseteq \mathcal{D}$ by the definition of $\mathcal{G}$. Since $\mathcal{A}$ generates $\mathcal{S}$, it follows that $\mathcal{D} = \mathcal{S}$. Therefore for every $B \in \mathcal{S}$, $\{X \in B\} \in \mathcal{G}$, so

$$\sigma(X) \subseteq \mathcal{G}.$$

Thus $\sigma(X) = \sigma(X^{-1}(\mathcal{A}))$.

Exercise 1.3.2

Question. Prove Theorem 1.3.6 when $n = 2$ by checking directly that $\{X_1 + X_2 < x\} \in \mathcal{F}$.

Knowledge used. `durrett_ch1_measurable_arithmetic`; `durrett_ch1_measurability_generator`; `durrett_ch1_sigma_field`.

Solution. Let X_1 and X_2 be random variables. For fixed $x \in \mathbb{R}$, observe that

$$\{X_1 + X_2 < x\} = \bigcup_{q \in \mathbb{Q}} (\{X_1 < q\} \cap \{X_2 < x - q\}).$$

Indeed, if $X_1(\omega) + X_2(\omega) < x$, then there is a rational q with

$$X_1(\omega) < q < x - X_2(\omega).$$

Conversely, if $X_1(\omega) < q$ and $X_2(\omega) < x - q$, then $X_1(\omega) + X_2(\omega) < x$.

For each rational q , the events $\{X_1 < q\}$ and $\{X_2 < x - q\}$ are in $\mathcal{F}$. The rationals are countable, so the union above is a countable union of measurable events. Hence $\{X_1 + X_2 < x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. Therefore $X_1 + X_2$ is a random variable.

Exercise 1.3.3

Question. Show that if f is continuous and $X_n \rightarrow X$ almost surely, then $f(X_n) \rightarrow f(X)$ almost surely.

Knowledge used. `durrett_ch1_composition_measurable`; `durrett_ch1_limits_measurable`.

Solution. Let

$$\Omega_0 = \{\omega : X_n(\omega) \rightarrow X(\omega)\}.$$

By assumption, $P(\Omega_0) = 1$. If $\omega \in \Omega_0$, then $X_n(\omega) \rightarrow X(\omega)$ as real numbers. Since f is continuous,

$$f(X_n(\omega)) \rightarrow f(X(\omega)).$$

Thus convergence holds for every $\omega \in \Omega_0$, a probability-one set. Therefore $f(X_n) \rightarrow f(X)$ almost surely.

Exercise 1.3.4

Question.

- (i) Show that a continuous function from $\mathbb{R}^d$ to $\mathbb{R}$ is measurable from $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.
- (ii) Show that $\mathcal{B}(\mathbb{R}^d)$ is the smallest σ -field that makes all continuous functions measurable.

Knowledge used. `durrett_ch1_composition_measurable`;
`durrett_ch1_measurability_generator`; `durrett_ch1_sigma_field`.

Solution.

- (i) Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be continuous. If $G \subseteq \mathbb{R}$ is open, then $f^{-1}(G)$ is open in $\mathbb{R}^d$, hence belongs to $\mathcal{B}(\mathbb{R}^d)$. Since the Borel σ -field on $\mathbb{R}$ is generated by open sets, f is Borel measurable.
- (ii) Let $\mathcal{G}$ be a σ -field on $\mathbb{R}^d$ that makes every continuous real-valued function measurable. We show $\mathcal{B}(\mathbb{R}^d) \subseteq \mathcal{G}$.

For a closed set $C \subseteq \mathbb{R}^d$, define

$$d_C(x) = \inf_{y \in C} |x - y|.$$

The function d_C is continuous, so it is $\mathcal{G}$ -measurable. Since

$$C = \{x : d_C(x) = 0\} = d_C^{-1}(\{0\}),$$

we have $C \in \mathcal{G}$. Thus every closed set belongs to $\mathcal{G}$, and therefore every open set belongs to $\mathcal{G}$ as well. Hence $\mathcal{B}(\mathbb{R}^d) \subseteq \mathcal{G}$.

On the other hand, part (i) shows that $\mathcal{B}(\mathbb{R}^d)$ itself makes all continuous functions measurable. Therefore $\mathcal{B}(\mathbb{R}^d)$ is the smallest such σ -field.

Exercise 1.3.5

Question. A function f is lower semicontinuous if

$$\liminf_{y \rightarrow x} f(y) \geq f(x),$$

and upper semicontinuous if $-f$ is lower semicontinuous. Show that f is lower semicontinuous if and only if $\{x : f(x) \leq a\}$ is closed for each $a \in \mathbb{R}$, and conclude that semicontinuous functions are measurable.

Knowledge used. `durrett_ch1_measurability_generator`; `durrett_ch1_limits_measurable`; `durrett_ch1_sigma_field`.

Solution. Suppose first that f is lower semicontinuous. Let $x_n \in \{x : f(x) \leq a\}$ and $x_n \rightarrow x$. Then

$$f(x) \leq \liminf_{n \rightarrow \infty} f(x_n) \leq a.$$

Hence $x \in \{f \leq a\}$, so $\{f \leq a\}$ is closed.

Conversely, assume that $\{f \leq a\}$ is closed for every $a \in \mathbb{R}$. Fix x and take any sequence $y_n \rightarrow x$. If $\liminf_n f(y_n) < f(x)$, choose a such that

$$\liminf_n f(y_n) < a < f(x).$$

Then infinitely many y_n lie in the closed set $\{f \leq a\}$, and a subsequence of those points converges to x . Closedness gives $x \in \{f \leq a\}$, contradicting $a < f(x)$. Therefore

$$\liminf_{y \rightarrow x} f(y) \geq f(x),$$

so f is lower semicontinuous.

If f is lower semicontinuous, then each set $\{f \leq a\}$ is closed and hence Borel. This implies measurability because, for example,

$$\{f < a\} = \bigcup_{n=1}^{\infty} \{f \leq a - 1/n\}$$

is Borel. If f is upper semicontinuous, then $-f$ is lower semicontinuous and measurable, so f is measurable as well.

Exercise 1.3.6

Question. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be arbitrary and define

$$f^\delta(x) = \sup\{f(y) : |y - x| < \delta\}, \quad f_\delta(x) = \inf\{f(y) : |y - x| < \delta\}.$$

Show that f^δ is lower semicontinuous and f_δ is upper semicontinuous. Let

$$f^0 = \lim_{\delta \downarrow 0} f^\delta, \quad f_0 = \lim_{\delta \downarrow 0} f_\delta.$$

Conclude that the set of discontinuities of f is $\{f^0 \neq f_0\}$ and is measurable.

Knowledge used. `durrett_ch1_limits_measurable`; `durrett_ch1_measurability_generator`; `durrett_ch1_composition_measurable`.

Solution. We first prove that f^δ is lower semicontinuous. It is enough to show that $\{x : f^\delta(x) > a\}$ is open for each $a \in \mathbb{R}$. If $f^\delta(x) > a$, then there is y with $|y - x| < \delta$ and $f(y) > a$. For all x' sufficiently close to x , we still have $|y - x'| < \delta$, so $f^\delta(x') > a$. Hence $\{f^\delta > a\}$ is open, and f^δ is lower semicontinuous.

Similarly,

$$f_\delta(x) = -\sup\{-f(y) : |y - x| < \delta\},$$

so f_δ is upper semicontinuous.

As $\delta \downarrow 0$, the quantities $f^\delta(x)$ decrease and $f_\delta(x)$ increase, so f^0 and f_0 are pointwise limits of measurable functions. Hence f^0 and f_0 are measurable.

For every x and every $\delta > 0$,

$$f_\delta(x) \leq f(x) \leq f^\delta(x),$$

so $f_0(x) \leq f(x) \leq f^0(x)$. If f is continuous at x , then the oscillation of f on small balls around x tends to 0, and therefore $f^0(x) = f_0(x) = f(x)$. Conversely, if $f^0(x) = f_0(x)$, then the upper and lower values of f on small balls squeeze to the same number, which must be $f(x)$; hence f is continuous at x . Therefore the discontinuity set is exactly

$$\{x : f^0(x) \neq f_0(x)\}.$$

Since f^0 and f_0 are measurable, this set is measurable.

Exercise 1.3.7

Question. A function $\phi : \Omega \rightarrow \mathbb{R}$ is simple if

$$\phi(\omega) = \sum_{m=1}^n c_m \mathbf{1}_{A_m}(\omega), \quad A_m \in \mathcal{F}.$$

Show that the class of $\mathcal{F}$ -measurable functions is the smallest class containing the simple functions and closed under pointwise limits.

Knowledge used. [durrett_ch1_simple_functions](#); [durrett_ch1_limits_measurable](#); [durrett_ch1_measurable_map](#).

Solution. Every simple function is measurable, since for each Borel set B ,

$$\{\phi \in B\} = \bigcup_{m: c_m \in B} A_m,$$

a finite union of measurable sets. Also, pointwise limits of measurable functions are measurable. Indeed, if $f_n \rightarrow f$, then

$$f = \limsup_{n \rightarrow \infty} f_n,$$

and limits, limsups, and liminfs of measurable functions are measurable. Thus the class of measurable functions contains the simple functions and is closed under pointwise limits.

It remains to show minimality. Let $\mathcal{H}$ be any class of functions that contains the simple functions and is closed under pointwise limits. If $f \geq 0$ is measurable, define

$$f_n(\omega) = 2^{-n} \lfloor 2^n (f(\omega) \wedge n) \rfloor.$$

Each f_n is a simple measurable function, and $f_n(\omega) \rightarrow f(\omega)$ for every ω . Hence $f \in \mathcal{H}$.

For a general real-valued measurable f , write $f = f^+ - f^-$. Approximate f^+ and f^- by simple functions and take their differences, which are again simple functions. This gives a sequence of simple functions converging pointwise to f . Therefore every measurable function belongs to $\mathcal{H}$. The measurable functions are consequently the smallest class containing the simple functions and closed under pointwise limits.

Exercise 1.3.8

Question. Use the previous exercise to conclude that Y is measurable with respect to $\sigma(X)$ if and only if

$$Y = f(X)$$

for some measurable function $f : \mathbb{R} \rightarrow \mathbb{R}$.

Knowledge used. [durrett_ch1_sigma_x_factorization](#); [durrett_ch1_composition_measurable](#); [durrett_ch1_simple_functions](#).

Solution. If $Y = f(X)$ for a Borel measurable f , then Y is $\sigma(X)$ -measurable because

$$\{Y \in B\} = \{f(X) \in B\} = \{X \in f^{-1}(B)\} \in \sigma(X).$$

Conversely, suppose Y is $\sigma(X)$ -measurable. Let $\mathcal{H}$ be the class of $\sigma(X)$ -measurable functions that can be written as $f(X)$ for some Borel measurable f . This class contains all $\sigma(X)$ -measurable simple functions: if

$$Y = \sum_{m=1}^n c_m \mathbf{1}_{A_m} \quad \text{and} \quad A_m \in \sigma(X),$$

then each $A_m = \{X \in B_m\}$ for some Borel set B_m , and

$$Y = \left(\sum_{m=1}^n c_m \mathbf{1}_{B_m} \right) (X).$$

The class $\mathcal{H}$ is also closed under pointwise limits, because if $Y_n = f_n(X)$ and $Y_n \rightarrow Y$, then

$$Y = f(X),$$

where f is defined as the pointwise limit of f_n on the Borel set where that real limit exists, and is defined to be 0 elsewhere. This f is Borel measurable because pointwise limsup and liminf are measurable. By Exercise 1.3.7, every $\sigma(X)$ -measurable real-valued function belongs to $\mathcal{H}$. Thus $Y = f(X)$ for some Borel measurable f .

Exercise 1.3.9

Question. Give a constructive proof of the previous result. Note that

$$\{\omega : m2^{-n} \leq Y(\omega) < (m+1)2^{-n}\} = \{X \in B_{m,n}\}$$

for some Borel set $B_{m,n}$, and define $f_n(x) = m2^{-n}$ for $x \in B_{m,n}$. Show that $f_n(x) \rightarrow f(x)$ and $Y = f(X)$.

Knowledge used. [durrett_ch1_sigma_x_factorization](#); [durrett_ch1_measurability_generator](#); [durrett_ch1_limits_measurable](#).

Solution. Assume Y is $\sigma(X)$ -measurable. For each $m \in \mathbb{Z}$ and $n \geq 1$, let

$$A_{m,n} = \{\omega : m2^{-n} \leq Y(\omega) < (m+1)2^{-n}\}.$$

Then $A_{m,n} \in \sigma(X)$, so by the definition of $\sigma(X)$ there is a Borel set $B_{m,n} \subseteq \mathbb{R}$ such that

$$A_{m,n} = \{X \in B_{m,n}\}.$$

For fixed n , the sets $A_{m,n}$ partition Ω . The corresponding Borel sets $B_{m,n}$ may overlap away from the actual range of X , so we harmlessly make them disjoint. Choose an enumeration $m_1, m_2, \dots$ of $\mathbb{Z}$ and set

$$C_{m_j,n} = B_{m_j,n} \setminus \bigcup_{i < j} B_{m_i,n}.$$

Then the $C_{m,n}$ are Borel and disjoint, and still

$$A_{m,n} = \{X \in C_{m,n}\}.$$

Indeed, points removed from $B_{m,n}$ cannot be hit by $X(\omega)$ with $\omega \in A_{m,n}$, since that would put ω in two different partition elements.

Define

$$f_n(x) = \sum_{m \in \mathbb{Z}} m2^{-n} \mathbf{1}_{C_{m,n}}(x),$$

with value 0 on the complement of $\bigcup_m C_{m,n}$. Each f_n is Borel measurable. Moreover, if $\omega \in A_{m,n}$, then

$$f_n(X(\omega)) = m2^{-n} \quad \text{and} \quad m2^{-n} \leq Y(\omega) < (m+1)2^{-n}.$$

Thus

$$0 \leq Y(\omega) - f_n(X(\omega)) < 2^{-n}$$

for every ω . Hence $f_n(X) \rightarrow Y$ pointwise.

Now let D be the Borel set on which the real limit $\lim_{n \rightarrow \infty} f_n(x)$ exists, and define

$$f(x) = \begin{cases} \lim_{n \rightarrow \infty} f_n(x), & x \in D, \\ 0, & x \notin D. \end{cases}$$

The set D is Borel because it is described by equality of the measurable functions $\limsup_n f_n$ and $\liminf_n f_n$, and f is Borel measurable. Since $f_n(X(\omega))$ converges for every ω , we have $X(\omega) \in D$. Evaluating at $X(\omega)$ gives

$$f(X(\omega)) = \lim_{n \rightarrow \infty} f_n(X(\omega)) = Y(\omega).$$

Therefore $Y = f(X)$, giving the desired constructive representation.

Exercise 1.4.1

Question. Show that if $f \geq 0$ and

$$\int f \, d\mu = 0,$$

then $f = 0$ almost everywhere.

Knowledge used. [durrett_ch1_nonnegative_integral](#); [durrett_ch1_integral_linearity](#); [durrett_ch1_measure_continuity](#).

Solution. For each $n \geq 1$, let

$$A_n = \{x : f(x) > 1/n\}.$$

On A_n we have $f \geq (1/n)\mathbf{1}_{A_n}$, so by monotonicity of the integral,

$$0 = \int f \, d\mu \geq \int \frac{1}{n} \mathbf{1}_{A_n} \, d\mu = \frac{1}{n} \mu(A_n).$$

Hence $\mu(A_n) = 0$ for every n .

Now

$$\{x : f(x) > 0\} = \bigcup_{n=1}^{\infty} A_n.$$

This is a countable union of null sets, so it has measure zero. Thus $f = 0$ almost everywhere.

Exercise 1.4.2

Question. Let $f \geq 0$ and

$$E_{n,m} = \left\{ x : \frac{m}{2^n} \leq f(x) < \frac{m+1}{2^n} \right\}.$$

Show that, as $n \rightarrow \infty$,

$$\sum_{m=1}^{\infty} \frac{m}{2^n} \mu(E_{n,m}) \uparrow \int f \, d\mu.$$

Knowledge used. [durrett_ch1_simple_functions](#); [durrett_ch1_nonnegative_integral](#); [durrett_ch1_monotone_convergence](#).

Solution. Define the dyadic lower approximation

$$\phi_n(x) = \sum_{m=1}^{\infty} \frac{m}{2^n} \mathbf{1}_{E_{n,m}}(x).$$

Equivalently,

$$\phi_n(x) = 2^{-n} \lfloor 2^n f(x) \rfloor$$

when $f(x) < \infty$, with the natural interpretation for extended values. Thus

$$0 \leq \phi_n(x) \leq f(x) \quad \text{and} \quad \phi_n(x) \uparrow f(x)$$

pointwise as $n \rightarrow \infty$.

The integral of ϕ_n is exactly

$$\int \phi_n \, d\mu = \sum_{m=1}^{\infty} \frac{m}{2^n} \mu(E_{n,m}),$$

where the $m = 0$ term would contribute 0 and is therefore omitted. Since $\phi_n \uparrow f$, the defining monotone approximation property of the nonnegative integral gives

$$\int \phi_n \, d\mu \uparrow \int f \, d\mu.$$

Hence the displayed sums increase to $\int f \, d\mu$.

Exercise 1.4.3

Question. Let g be an integrable function on $\mathbb{R}$ and let $\varepsilon > 0$. Show, using the definition of the integral and interval approximation of measurable sets, that g can be approximated in L^1 by a continuous function. More explicitly:

- (i) approximate g by a simple function ϕ with $\int |g - \phi| dx < \varepsilon$;
- (ii) approximate the sets in ϕ by finite unions of intervals to obtain a step function q with $\int |\phi - q| dx < \varepsilon$;
- (iii) round the corners of q to obtain a continuous function r with $\int |q - r| dx < \varepsilon$.

Knowledge used. `durrett_ch1_integrable_function`; `durrett_ch1_simple_functions`; `durrett_ch1_integral_linearity`.

Solution. We prove the three approximations in order.

- (i) Since g is integrable, by the construction of the Lebesgue integral there is a simple function

$$\phi = \sum_{k=1}^N b_k \mathbf{1}_{A_k}$$

such that

$$\int_{\mathbb{R}} |g - \phi| dx < \varepsilon.$$

One way to see this is to approximate g^+ and g^- from below by simple functions and then subtract the approximations.

- (ii) By the interval approximation result for Lebesgue measurable sets, each A_k can be approximated in measure by a finite union U_k of intervals so well that

$$\sum_{k=1}^N |b_k| \lambda(A_k \triangle U_k) < \varepsilon,$$

where λ denotes Lebesgue measure. Put

$$\psi = \sum_{k=1}^N b_k \mathbf{1}_{U_k}.$$

Then

$$\int_{\mathbb{R}} |\phi - \psi| dx \leq \sum_{k=1}^N |b_k| \lambda(A_k \triangle U_k) < \varepsilon.$$

Refining the finitely many interval endpoints, ψ can be rewritten as a step function

$$q = \sum_{j=1}^M c_j \mathbf{1}_{(a_{j-1}, a_j)}$$

with $a_0 < a_1 < \dots < a_M$ and

$$\int_{\mathbb{R}} |\phi - q| dx < \varepsilon.$$

(iii) For each interval (a_{j-1}, a_j) choose $\delta_j > 0$ so small that $2\delta_j < a_j - a_{j-1}$ and

$$\sum_{j=1}^M 2|c_j|\delta_j < \varepsilon.$$

Define r_j to be continuous, equal to 0 outside (a_{j-1}, a_j) , equal to c_j on $[a_{j-1} + \delta_j, a_j - \delta_j]$, and linear on the two short boundary intervals. Let

$$r = \sum_{j=1}^M r_j.$$

Then r is continuous and differs from q only on boundary intervals whose total weighted length is controlled by the choice of the δ_j . Hence

$$\int_{\mathbb{R}} |q - r| dx \leq \sum_{j=1}^M 2|c_j|\delta_j < \varepsilon.$$

Combining the three estimates,

$$\int_{\mathbb{R}} |g - r| dx \leq \int_{\mathbb{R}} |g - \phi| dx + \int_{\mathbb{R}} |\phi - q| dx + \int_{\mathbb{R}} |q - r| dx < 3\varepsilon.$$

Since ε was arbitrary, this proves that integrable functions can be approximated in L^1 by continuous functions.

Exercise 1.4.4

Question. Prove the Riemann–Lebesgue lemma: if g is integrable, then

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} g(x) \cos(nx) dx = 0.$$

Hint: first prove it for step functions, then use the previous exercise.

Knowledge used. [durrett_ch1_integrable_function](#); [durrett_ch1_integral_linearity](#); [durrett_ch1_dominated_convergence](#).

Solution. First suppose q is a step function:

$$q(x) = \sum_{j=1}^M c_j \mathbf{1}_{(a_{j-1}, a_j)}(x).$$

Then

$$\int_{\mathbb{R}} q(x) \cos(nx) dx = \sum_{j=1}^M c_j \int_{a_{j-1}}^{a_j} \cos(nx) dx = \sum_{j=1}^M c_j \frac{\sin(na_j) - \sin(na_{j-1})}{n}.$$

Each term is $O(1/n)$, so this tends to 0 as $n \rightarrow \infty$.

Now let $g \in L^1(\mathbb{R})$ and fix $\varepsilon > 0$. By Exercise 1.4.3, or already by its step-function approximation stage, choose a step function q such that

$$\int_{\mathbb{R}} |g - q| dx < \varepsilon.$$

Then

$$\begin{aligned} \left| \int_{\mathbb{R}} g(x) \cos(nx) dx \right| &\leq \left| \int_{\mathbb{R}} q(x) \cos(nx) dx \right| + \int_{\mathbb{R}} |g(x) - q(x)| |\cos(nx)| dx \\ &\leq \left| \int_{\mathbb{R}} q(x) \cos(nx) dx \right| + \varepsilon. \end{aligned}$$

Taking $\limsup_{n \rightarrow \infty}$ and using the step-function case gives

$$\limsup_{n \rightarrow \infty} \left| \int_{\mathbb{R}} g(x) \cos(nx) dx \right| \leq \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, the limit is 0.

Exercise 1.5.1

Question. Let

$$\|f\|_{\infty} = \inf\{M : \mu(\{x : |f(x)| > M\}) = 0\}.$$

Prove that

$$\int |fg| d\mu \leq \|f\|_1 \|g\|_{\infty}.$$

Knowledge used. [durrett_ch1_holder_integral](#); [durrett_ch1_integrable_function](#); [durrett_ch1_integral_linearity](#).

Solution. Let $M = \|g\|_{\infty}$. By definition of essential supremum, $|g| \leq M$ almost everywhere. Therefore

$$|fg| \leq M|f|$$

almost everywhere. By monotonicity and homogeneity of the integral,

$$\int |fg| d\mu \leq \int M|f| d\mu = M \int |f| d\mu = \|g\|_{\infty} \|f\|_1.$$

This proves the desired inequality.

Exercise 1.5.2

Question. Show that if μ is a probability measure, then

$$\|f\|_{\infty} = \lim_{p \rightarrow \infty} \|f\|_p.$$

Knowledge used. [durrett_ch1_jensen_integral](#); [durrett_ch1_holder_integral](#); [durrett_ch1_monotone_convergence](#).

Solution. Let $M = \|f\|_{\infty}$. First assume $M < \infty$. Since $|f| \leq M$ almost everywhere and $\mu(\Omega) = 1$,

$$\|f\|_p^p = \int |f|^p d\mu \leq M^p,$$

so $\|f\|_p \leq M$ for all p .

For the reverse inequality, fix $\varepsilon > 0$ and let

$$A_{\varepsilon} = \{|f| > M - \varepsilon\}.$$

By the definition of essential supremum, $\mu(A_\varepsilon) > 0$. Hence

$$\|f\|_p^p \geq \int_{A_\varepsilon} |f|^p d\mu \geq (M - \varepsilon)^p \mu(A_\varepsilon).$$

Taking p th roots,

$$\|f\|_p \geq (M - \varepsilon) \mu(A_\varepsilon)^{1/p}.$$

Letting $p \rightarrow \infty$ gives

$$\liminf_{p \rightarrow \infty} \|f\|_p \geq M - \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary and $\limsup_p \|f\|_p \leq M$, we obtain $\|f\|_p \rightarrow M$.

If $M = \infty$, then for every $K > 0$, the set $\{|f| > K\}$ has positive measure. The same lower bound gives

$$\liminf_{p \rightarrow \infty} \|f\|_p \geq K.$$

Since K is arbitrary, $\|f\|_p \rightarrow \infty$. Thus the statement holds also in the infinite case.

Exercise 1.5.3

Question. Prove Minkowski's inequality. For $p \in (1, \infty)$, use Holder's inequality to show

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Then show the same result for $p = 1$ and $p = \infty$.

Knowledge used. [durrett_ch1_minkowski](#); [durrett_ch1_holder_integral](#); [durrett_ch1_integral_linearity](#).

Solution. Let $1 < p < \infty$ and let q be the conjugate exponent: $1/p + 1/q = 1$. If $\|f + g\|_p = 0$, there is nothing to prove. Otherwise,

$$\|f + g\|_p^p = \int |f + g|^p d\mu \leq \int |f| |f + g|^{p-1} d\mu + \int |g| |f + g|^{p-1} d\mu.$$

By Holder's inequality,

$$\int |f| |f + g|^{p-1} d\mu \leq \|f\|_p \left(\int |f + g|^{(p-1)q} d\mu \right)^{1/q}.$$

Since $(p-1)q = p$, the last factor is

$$\left(\int |f + g|^p d\mu \right)^{1/q} = \|f + g\|_p^{p/q} = \|f + g\|_p^{p-1}.$$

The same estimate holds with g in place of f . Therefore

$$\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p) \|f + g\|_p^{p-1}.$$

Dividing by $\|f + g\|_p^{p-1}$ gives Minkowski's inequality.

For $p = 1$,

$$\|f + g\|_1 = \int |f + g| d\mu \leq \int |f| d\mu + \int |g| d\mu = \|f\|_1 + \|g\|_1.$$

For $p = \infty$, $|f + g| \leq |f| + |g|$ almost everywhere, hence

$$\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty.$$

Exercise 1.5.4

Question. If f is integrable and E_m are disjoint sets with union E , show that

$$\sum_{m=0}^{\infty} \int_{E_m} f \, d\mu = \int_E f \, d\mu.$$

Conclude that if $f \geq 0$, then

$$\nu(E) = \int_E f \, d\mu$$

defines a measure.

Knowledge used. `durrett_ch1_integrable_function`; `durrett_ch1_integral_linearity`; `durrett_ch1_monotone_convergence`.

Solution. First suppose $f \geq 0$. Since the sets E_m are disjoint,

$$\mathbf{1}_E f = \sum_{m=0}^{\infty} \mathbf{1}_{E_m} f.$$

Let

$$s_n = \sum_{m=0}^n \mathbf{1}_{E_m} f.$$

Then $s_n \uparrow \mathbf{1}_E f$. By monotone convergence,

$$\int_E f \, d\mu = \int \mathbf{1}_E f \, d\mu = \lim_{n \rightarrow \infty} \int s_n \, d\mu = \lim_{n \rightarrow \infty} \sum_{m=0}^n \int_{E_m} f \, d\mu.$$

For a general integrable f , write $f = f^+ - f^-$. Applying the nonnegative case to f^+ and f^- and subtracting gives the desired identity.

If $f \geq 0$ and $\nu(E) = \int_E f \, d\mu$, then $\nu(\emptyset) = 0$ and $\nu(E) \geq 0$. The identity above gives countable additivity over disjoint unions, so ν is a measure.

Exercise 1.5.5

Question. If $g_n \uparrow g$ and

$$\int g_1^- \, d\mu < \infty,$$

show that

$$\int g_n \, d\mu \uparrow \int g \, d\mu.$$

Knowledge used. `durrett_ch1_monotone_convergence`; `durrett_ch1_integrable_function`; `durrett_ch1_integral_linearity`.

Solution. Since $g_n \uparrow g$, the functions

$$h_n = g_n - g_1$$

are nonnegative and increase pointwise to

$$h = g - g_1.$$

By monotone convergence,

$$\int h_n d\mu \uparrow \int h d\mu.$$

Since $\int g_1^- d\mu < \infty$, the integrals involving g_1 are well-defined in the extended sense, and

$$\int g_n d\mu = \int g_1 d\mu + \int h_n d\mu \uparrow \int g_1 d\mu + \int h d\mu = \int g d\mu.$$

Exercise 1.5.6

Question. If $g_m \geq 0$, show that

$$\int \sum_{m=0}^{\infty} g_m d\mu = \sum_{m=0}^{\infty} \int g_m d\mu.$$

Knowledge used. [durrett_ch1_monotone_convergence](#); [durrett_ch1_nonnegative_integral](#); [durrett_ch1_integral_linearity](#).

Solution. Let

$$s_n = \sum_{m=0}^n g_m.$$

Then $s_n \geq 0$ and

$$s_n \uparrow \sum_{m=0}^{\infty} g_m.$$

By monotone convergence,

$$\int \sum_{m=0}^{\infty} g_m d\mu = \lim_{n \rightarrow \infty} \int s_n d\mu.$$

For each finite n , linearity gives

$$\int s_n d\mu = \sum_{m=0}^n \int g_m d\mu.$$

Taking $n \rightarrow \infty$ proves the formula.

Exercise 1.5.7

Question. Let $f \geq 0$.

(i) Show that

$$\int f \wedge n d\mu \uparrow \int f d\mu.$$

(ii) Use this to conclude that if g is integrable and $\varepsilon > 0$, then there is $\delta > 0$ such that $\mu(A) < \delta$ implies

$$\int_A |g| d\mu < \varepsilon.$$

Knowledge used. [durrett_ch1_monotone_convergence](#); [durrett_ch1_integrable_function](#); [durrett_ch1_integral_linearity](#).

Solution.

(i) Since $f \wedge n \uparrow f$, monotone convergence gives

$$\int f \wedge n \, d\mu \uparrow \int f \, d\mu.$$

(ii) Apply part (i) to $f = |g|$. Since g is integrable,

$$\int (|g| - |g| \wedge n) \, d\mu \downarrow 0.$$

Choose n such that

$$\int (|g| - |g| \wedge n) \, d\mu < \varepsilon/2.$$

If A is measurable, then

$$\int_A |g| \, d\mu \leq \int_A (|g| \wedge n) \, d\mu + \int (|g| - |g| \wedge n) \, d\mu \leq n\mu(A) + \varepsilon/2.$$

Now choose $\delta = \varepsilon/(2n)$, with any positive δ if $n = 0$. Then $\mu(A) < \delta$ implies

$$\int_A |g| \, d\mu < \varepsilon.$$

Exercise 1.5.8

Question. Show that if f is integrable on $[a, b]$ and

$$g(x) = \int_{[a, x]} f(y) \, dy,$$

then g is continuous on (a, b) .

Knowledge used. [durrett_ch1_integrable_function](#); [durrett_ch1_integral_linearity](#); [durrett_ch1_dominated_convergence](#).

Solution. Fix $x \in (a, b)$ and let $\varepsilon > 0$. By Exercise 1.5.7, there is $\delta > 0$ such that whenever a measurable set A has Lebesgue measure less than δ ,

$$\int_A |f(y)| \, dy < \varepsilon.$$

If h is small enough that $x + h \in (a, b)$ and $|h| < \delta$, then

$$g(x + h) - g(x) = \begin{cases} \int_{(x, x+h]} f(y) \, dy, & h > 0, \\ -\int_{(x+h, x]} f(y) \, dy, & h < 0. \end{cases}$$

In either case,

$$|g(x + h) - g(x)| \leq \int_{(x \wedge (x+h), x \vee (x+h)]} |f(y)| \, dy < \varepsilon.$$

Thus $g(x + h) \rightarrow g(x)$ as $h \rightarrow 0$, so g is continuous on (a, b) .

Exercise 1.5.9

Question. Show that if

$$\|f\|_p = \left(\int |f|^p d\mu \right)^{1/p} < \infty,$$

then there are simple functions ϕ_n such that

$$\|\phi_n - f\|_p \rightarrow 0.$$

Knowledge used. [durrett_ch1_simple_functions](#); [durrett_ch1_dominated_convergence](#); [durrett_ch1_integrable_function](#).

Solution. Assume $1 \leq p < \infty$. Since f is measurable, there exists a sequence of simple functions ϕ_n such that

$$\phi_n(x) \rightarrow f(x) \quad \text{and} \quad |\phi_n(x)| \leq |f(x)|$$

for all x ; construct this by applying dyadic simple approximations to f^+ and f^- .

Then

$$|\phi_n - f|^p \leq (|\phi_n| + |f|)^p \leq (2|f|)^p = 2^p |f|^p.$$

The function $2^p |f|^p$ is integrable because $\|f\|_p < \infty$. Also $|\phi_n - f|^p \rightarrow 0$ pointwise. By dominated convergence,

$$\int |\phi_n - f|^p d\mu \rightarrow 0.$$

Therefore $\|\phi_n - f\|_p \rightarrow 0$.

Exercise 1.5.10

Question. Show that if

$$\sum_n \int |f_n| d\mu < \infty,$$

then

$$\sum_n \int f_n d\mu = \int \sum_n f_n d\mu.$$

Knowledge used. [durrett_ch1_dominated_convergence](#); [durrett_ch1_monotone_convergence](#); [durrett_ch1_integral_linearity](#).

Solution. By Exercise 1.5.6 applied to the nonnegative functions $|f_n|$,

$$\int \sum_n |f_n| d\mu = \sum_n \int |f_n| d\mu < \infty.$$

Hence

$$H = \sum_n |f_n|$$

is integrable and finite almost everywhere. Therefore

$$\sum_n f_n$$

converges absolutely almost everywhere.

Let

$$S_N = \sum_{n=0}^N f_n.$$

Then $S_N \rightarrow S = \sum_n f_n$ almost everywhere and

$$|S_N| \leq H$$

for all N . By dominated convergence,

$$\int S_N d\mu \rightarrow \int S d\mu.$$

For finite sums, linearity gives

$$\int S_N d\mu = \sum_{n=0}^N \int f_n d\mu.$$

Letting $N \rightarrow \infty$ yields

$$\int \sum_n f_n d\mu = \sum_n \int f_n d\mu.$$

Exercise 1.6.1

Question. Suppose ϕ is strictly convex. Under the assumptions of Jensen's inequality, show that

$$\phi(EX) = E\phi(X)$$

implies $X = EX$ almost surely.

Knowledge used. [durrett_ch1_jensen_integral](#); [durrett_ch1_expectation_as_integral](#).

Solution. Let $m = EX$. If $P(X = m) = 1$, equality is clear. Conversely suppose $P(X \neq m) > 0$. Then there are numbers $a < m < b$ such that

$$P(X \leq a) > 0, \quad P(X \geq b) > 0.$$

By strict convexity, the graph of ϕ lies strictly above any supporting chord except at the endpoints. Equivalently, Jensen's inequality is strict for a nonconstant random variable. More explicitly, condition on the two sets $\{X < m\}$ and $\{X \geq m\}$ and apply strict convexity to the two conditional means; the inequality is strict because both sets have positive probability and their conditional means are different. Thus

$$E\phi(X) > \phi(EX),$$

contradicting equality. Hence $P(X = EX) = 1$.

Exercise 1.6.2

Question. Suppose $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex. Show that

$$E\phi(X_1, \dots, X_n) \geq \phi(EX_1, \dots, EX_n),$$

provided $E|\phi(X_1, \dots, X_n)| < \infty$ and $E|X_i| < \infty$ for all i .

Knowledge used. [durrett_ch1_jensen_integral](#); [durrett_ch1_expectation_linearity](#).

Solution. Let

$$m = (EX_1, \dots, EX_n).$$

For a convex function on $\mathbb{R}^n$, there is a supporting hyperplane at m : there are $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$ such that

$$\phi(x) \geq a \cdot x + b \quad \text{for all } x, \quad \phi(m) = a \cdot m + b.$$

Therefore

$$E\phi(X_1, \dots, X_n) \geq E[a \cdot (X_1, \dots, X_n) + b] = a \cdot m + b = \phi(m).$$

This proves the multidimensional Jensen inequality.

Exercise 1.6.3

Question. Chebyshev's inequality is and is not sharp.

- (i) Show that if $0 < b \leq a$ are fixed, there is X with $EX^2 = b^2$ for which $P(|X| \geq a) = b^2/a^2$.
- (ii) Show that if $0 < EX^2 < \infty$, then

$$\lim_{a \rightarrow \infty} \frac{a^2 P(|X| \geq a)}{EX^2} = 0.$$

Knowledge used. [durrett_ch1_markov_chebyshev](#); [durrett_ch1_expectation_limit_theorems](#).

Solution.

- (i) Let

$$P(X = a) = \frac{b^2}{2a^2}, \quad P(X = -a) = \frac{b^2}{2a^2}, \quad P(X = 0) = 1 - \frac{b^2}{a^2}.$$

Then

$$EX^2 = a^2 \frac{b^2}{a^2} = b^2 \quad \text{and} \quad P(|X| \geq a) = \frac{b^2}{a^2}.$$

So Chebyshev's bound can be attained at a fixed level a .

- (ii) Let $Y = X^2$. Since $EY < \infty$,

$$a^2 P(|X| \geq a) = a^2 P(Y \geq a^2) \leq E(Y; Y \geq a^2).$$

As $a \rightarrow \infty$, $Y \mathbf{1}_{\{Y \geq a^2\}} \downarrow 0$ and is bounded by the integrable random variable Y . By dominated convergence,

$$E(Y; Y \geq a^2) \rightarrow 0.$$

Dividing by the positive constant EX^2 gives the desired limit.

Exercise 1.6.4

Question. One-sided Chebyshev bound.

- (i) Let $a > b > 0$, $0 < p < 1$, and let X satisfy $P(X = a) = p$, $P(X = -b) = 1 - p$. Apply Chebyshev's inequality to $\phi(x) = (x + b)^2$ and conclude that if Y has the same mean and variance as X , then $P(Y \geq a) \leq p$, with equality when $Y = X$.

(ii) If $EY = 0$, $\text{var}(Y) = \sigma^2$, and $a > 0$, show

$$P(Y \geq a) \leq \frac{\sigma^2}{a^2 + \sigma^2},$$

and show equality is possible.

Knowledge used. [durrett_ch1_markov_chebyshev](#); [durrett_ch1_expectation_linearity](#); [durrett_ch1_common_distributions_expectations](#).

Solution.

(i) If X has the stated distribution, then

$$EX = pa - (1 - p)b, \quad E(X + b)^2 = p(a + b)^2.$$

For any Y with the same mean and variance as X , we also have $E(Y + b)^2 = E(X + b)^2 = p(a + b)^2$, because $E(Y + b)^2$ is determined by EY and EY^2 . On the event $\{Y \geq a\}$,

$$(Y + b)^2 \geq (a + b)^2.$$

Thus Chebyshev's inequality gives

$$(a + b)^2 P(Y \geq a) \leq E(Y + b)^2 = p(a + b)^2,$$

so $P(Y \geq a) \leq p$. For $Y = X$, equality holds.

(ii) Apply part (i) with a two-point random variable taking values a and $-b$ and having mean 0. The mean-zero condition gives

$$pa - (1 - p)b = 0, \quad p = \frac{b}{a + b}.$$

Its variance is

$$\sigma^2 = pa^2 + (1 - p)b^2 = ab.$$

Thus $b = \sigma^2/a$ and

$$p = \frac{\sigma^2/a}{a + \sigma^2/a} = \frac{\sigma^2}{a^2 + \sigma^2}.$$

The first part gives

$$P(Y \geq a) \leq \frac{\sigma^2}{a^2 + \sigma^2}.$$

Equality is attained by the two-point variable with $P(Y = a) = \sigma^2/(a^2 + \sigma^2)$ and $P(Y = -\sigma^2/a) = a^2/(a^2 + \sigma^2)$.

Exercise 1.6.5

Question. Show that the following lower bounds do not exist:

(i) If $\varepsilon > 0$,

$$\inf\{P(|X| > \varepsilon) : EX = 0, \text{var}(X) = 1\} = 0.$$

(ii) If $y \geq 1$ and $\sigma^2 \in (0, \infty)$,

$$\inf\{P(|X| > y) : EX = 1, \text{var}(X) = \sigma^2\} = 0.$$

Knowledge used. [durrett_ch1_common_distributions_expectations](#);
[durrett_ch1_markov_chebyshev](#).

Solution.

(i) Fix $\delta > 0$. Let

$$P(X = 0) = 1 - \delta, \quad P(X = \delta^{-1/2}) = P(X = -\delta^{-1/2}) = \delta/2.$$

Then $EX = 0$ and $EX^2 = 1$, so $\text{var}(X) = 1$. Also $P(|X| > \varepsilon) \leq \delta$ for all sufficiently small δ such that $\delta^{-1/2} > \varepsilon$. Since δ is arbitrary, the infimum is 0.

(ii) Fix $\delta > 0$ and let Z have mean 0 and variance 1 with $P(Z = 0) = 1 - \delta$ as in part (i). Put

$$X = 1 + \sigma Z.$$

Then $EX = 1$ and $\text{var}(X) = \sigma^2$. The event $\{|X| > y\}$ is contained in the event $\{Z \neq 0\}$ unless one of the rare nonzero values occurs; hence

$$P(|X| > y) \leq P(Z \neq 0) = \delta.$$

Letting $\delta \downarrow 0$ gives the infimum 0.

Exercise 1.6.6

Question. Let $Y \geq 0$ with $EY^2 < \infty$. Apply Cauchy–Schwarz to $Y\mathbf{1}_{\{Y>0\}}$ and conclude

$$P(Y > 0) \geq \frac{(EY)^2}{EY^2}.$$

Knowledge used. [durrett_ch1_holder_integral](#); [durrett_ch1_expectation_linearity](#).

Solution. Since $Y = Y\mathbf{1}_{\{Y>0\}}$,

$$EY = E(Y\mathbf{1}_{\{Y>0\}}).$$

By Cauchy–Schwarz,

$$(EY)^2 \leq E(Y^2) E(\mathbf{1}_{\{Y>0\}}^2) = EY^2 P(Y > 0).$$

If $EY^2 > 0$, division gives

$$P(Y > 0) \geq \frac{(EY)^2}{EY^2}.$$

If $EY^2 = 0$, then $Y = 0$ a.s. and the assertion is trivial.

Exercise 1.6.7

Question. Let $\Omega = (0, 1)$ with Lebesgue measure, let $\alpha \in (1, 2)$, and set

$$X_n = n^\alpha \mathbf{1}_{(1/(n+1), 1/n)}.$$

Show $X_n \rightarrow 0$ almost surely and Theorem 1.6.8 applies with $h(x) = x$ and $g(x) = |x|^{2/\alpha}$, but the X_n are not dominated by an integrable random variable.

Knowledge used. [durrett_ch1_expectation_limit_theorems](#);
[durrett_ch1_dominated_convergence](#); [durrett_ch1_common_distributions_expectations](#).

Solution. The intervals $(1/(n+1), 1/n)$ are disjoint. Hence for each $\omega \in (0, 1)$, at most one $X_n(\omega)$ is nonzero. Thus $X_n(\omega) \rightarrow 0$ for every ω .

For $g(x) = |x|^{2/\alpha}$,

$$Eg(X_n) = n^2 \lambda(1/(n+1), 1/n) = n^2 \left(\frac{1}{n} - \frac{1}{n+1} \right) = \frac{n}{n+1} \leq 1.$$

Also $g(x) \rightarrow \infty$ as $|x| \rightarrow \infty$, and

$$\frac{|h(x)|}{g(x)} = |x|^{1-2/\alpha} \rightarrow 0 \quad \text{as } |x| \rightarrow \infty,$$

because $\alpha < 2$. Thus Theorem 1.6.8 applies.

Suppose, toward a contradiction, that $|X_n| \leq Y$ for all n with $EY < \infty$. Then on $(1/(n+1), 1/n)$, $Y \geq n^\alpha$. Therefore

$$EY \geq \sum_{n=1}^N n^\alpha \left(\frac{1}{n} - \frac{1}{n+1} \right) = \sum_{n=1}^N \frac{n^{\alpha-1}}{n+1}.$$

The terms are comparable to $n^{\alpha-2}$, and since $\alpha - 2 \in (-1, 0)$, the series diverges. This contradicts $EY < \infty$. Hence no integrable dominating random variable exists.

Exercise 1.6.8

Question. Suppose a probability measure μ has

$$\mu(A) = \int_A f(x) dx$$

for all Borel A . Show that for any g with $g \geq 0$ or $\int |g| d\mu < \infty$,

$$\int g(x) \mu(dx) = \int g(x) f(x) dx.$$

Knowledge used. [durrett_ch1_change_of_variables](#);
[durrett_ch1_expectation_limit_theorems](#).

Solution. First take $g = \mathbf{1}_A$. Then

$$\int \mathbf{1}_A d\mu = \mu(A) = \int_A f(x) dx = \int \mathbf{1}_A(x) f(x) dx.$$

By linearity, the result holds for nonnegative simple functions. If $g \geq 0$, choose simple $g_n \uparrow g$. Monotone convergence on both sides gives

$$\int g d\mu = \lim_n \int g_n d\mu = \lim_n \int g_n f dx = \int g f dx.$$

For integrable signed g , apply the nonnegative result to g^+ and g^- and subtract.

Exercise 1.6.9

Question. Let $A_1, \dots, A_n$ be events and $A = \bigcup_{i=1}^n A_i$. Prove the inclusion–exclusion formula.

Knowledge used. [durrett_ch1_inclusion_exclusion](#); [durrett_ch1_expectation_linearity](#).

Solution. For each ω ,

$$\mathbf{1}_A(\omega) = 1 - \prod_{i=1}^n (1 - \mathbf{1}_{A_i}(\omega)).$$

Indeed, the product is 1 exactly when none of the A_i occur, and is 0 otherwise. Expanding the product gives

$$\mathbf{1}_A = \sum_i \mathbf{1}_{A_i} - \sum_{i < j} \mathbf{1}_{A_i \cap A_j} + \sum_{i < j < k} \mathbf{1}_{A_i \cap A_j \cap A_k} - \dots + (-1)^{n-1} \mathbf{1}_{\bigcap_{i=1}^n A_i}.$$

Taking expectations yields

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n-1} P\left(\bigcap_{i=1}^n A_i\right).$$

Exercise 1.6.10

Question. Prove the Bonferroni inequalities for $A = \bigcup_{i=1}^n A_i$: stopping inclusion–exclusion after an even number of sums gives a lower bound, and stopping after an odd number of sums gives an upper bound.

Knowledge used. [durrett_ch1_inclusion_exclusion](#); [durrett_ch1_expectation_linearity](#).

Solution. For a fixed outcome ω , let $N(\omega) = \sum_i \mathbf{1}_{A_i}(\omega)$. Then $\mathbf{1}_A(\omega) = \mathbf{1}_{\{N(\omega) \geq 1\}}$. The inclusion–exclusion partial sums are the binomial sums

$$S_r(N) = \sum_{k=1}^r (-1)^{k-1} \binom{N}{k}.$$

For $N = 0$, all sums are 0. For $N \geq 1$, the full alternating sum is 1, and the alternating partial sums lie alternately above and below 1:

$$S_1(N) \geq 1, \quad S_2(N) \leq 1, \quad S_3(N) \geq 1, \quad \dots$$

Thus, pointwise,

$$\begin{aligned} \mathbf{1}_A &\leq \sum_i \mathbf{1}_{A_i}, \\ \mathbf{1}_A &\geq \sum_i \mathbf{1}_{A_i} - \sum_{i < j} \mathbf{1}_{A_i \cap A_j}, \end{aligned}$$

and

$$\mathbf{1}_A \leq \sum_i \mathbf{1}_{A_i} - \sum_{i < j} \mathbf{1}_{A_i \cap A_j} + \sum_{i < j < k} \mathbf{1}_{A_i \cap A_j \cap A_k}.$$

Taking expectations gives the displayed Bonferroni inequalities. The same pointwise alternating-sum argument gives the general even/lower and odd/upper rule.

Exercise 1.6.11

Question. If $E|X|^k < \infty$, show that for $0 < j < k$,

$$E|X|^j < \infty, \quad E|X|^j \leq (E|X|^k)^{j/k}.$$

Knowledge used. [durrett_ch1_jensen_integral](#); [durrett_ch1_holder_integral](#).

Solution. Let $Y = |X|^k \geq 0$ and let $r = j/k \in (0, 1)$. The function $t \mapsto t^r$ is concave on $[0, \infty)$. Jensen's inequality for the convex function $-t^r$ gives

$$EY^r \leq (EY)^r.$$

Since $Y^r = |X|^j$, we get

$$E|X|^j \leq (E|X|^k)^{j/k} < \infty.$$

Exercise 1.6.12

Question. Apply Jensen's inequality with $\phi(x) = e^x$ and $P(X = \log y_m) = p(m)$ to prove

$$\sum_{m=1}^n p(m)y_m \geq \prod_{m=1}^n y_m^{p(m)}$$

when $\sum_m p(m) = 1$ and $p(m), y_m > 0$.

Knowledge used. [durrett_ch1_jensen_integral](#);
[durrett_ch1_common_distributions_expectations](#).

Solution. Let X be the discrete random variable with $P(X = \log y_m) = p(m)$. Since $\phi(x) = e^x$ is convex, Jensen's inequality gives

$$E(e^X) \geq e^{EX}.$$

Now

$$E(e^X) = \sum_{m=1}^n p(m)y_m, \quad e^{EX} = \exp\left(\sum_{m=1}^n p(m)\log y_m\right) = \prod_{m=1}^n y_m^{p(m)}.$$

This proves the weighted arithmetic-geometric mean inequality. When $p(m) = 1/n$, it is the usual AM-GM inequality.

Exercise 1.6.13

Question. If $EX_1^- < \infty$ and $X_n \uparrow X$, show that

$$EX_n \uparrow EX.$$

Knowledge used. [durrett_ch1_expectation_limit_theorems](#);
[durrett_ch1_expectation_linearity](#).

Solution. Let

$$Y_n = X_n - X_1.$$

Then $Y_n \geq 0$ and $Y_n \uparrow X - X_1$. By monotone convergence,

$$EY_n \uparrow E(X - X_1).$$

Since $EX_1^- < \infty$, adding EX_1 is legitimate in the extended sense. Therefore

$$EX_n = EX_1 + EY_n \uparrow EX_1 + E(X - X_1) = EX.$$

Exercise 1.6.14

Question. Let $X \geq 0$, but do not assume $E(1/X) < \infty$. Show

$$\lim_{y \rightarrow \infty} yE(1/X; X > y) = 0, \quad \lim_{y \downarrow 0} yE(1/X; X > y) = 0.$$

Knowledge used. [durrett_ch1_tail_sum_nonnegative](#);
[durrett_ch1_expectation_limit_theorems](#).

Solution. For $y \rightarrow \infty$, on $\{X > y\}$ we have $y/X \leq 1$, so

$$0 \leq yE(1/X; X > y) = E(y/X; X > y) \leq P(X > y) \rightarrow 0.$$

For $y \downarrow 0$, fix $\eta > 0$. Choose $\delta > 0$ so small that

$$P(0 < X \leq \delta) < \eta,$$

which is possible by continuity of probability from above. If $0 < y < \delta$, then

$$\begin{aligned} yE(1/X; X > y) &= E(y/X; y < X \leq \delta) + E(y/X; X > \delta) \\ &\leq P(y < X \leq \delta) + \frac{y}{\delta} \leq \eta + \frac{y}{\delta}. \end{aligned}$$

Letting $y \downarrow 0$ gives a limsup at most η . Since η is arbitrary, the second limit is 0.

Exercise 1.6.15

Question. If $X_n \geq 0$, show that

$$E\left(\sum_{n=0}^{\infty} X_n\right) = \sum_{n=0}^{\infty} EX_n.$$

Knowledge used. [durrett_ch1_expectation_limit_theorems](#); [durrett_ch1_fubini_tonelli](#).

Solution. Let

$$S_N = \sum_{n=0}^N X_n.$$

Then $S_N \uparrow \sum_{n=0}^{\infty} X_n$. By monotone convergence,

$$E\left(\sum_{n=0}^{\infty} X_n\right) = \lim_{N \rightarrow \infty} ES_N.$$

For finite sums, expectation is linear, so

$$ES_N = \sum_{n=0}^N EX_n.$$

Letting $N \rightarrow \infty$ proves the identity.

Exercise 1.6.16

Question. If X is integrable and A_n are disjoint sets with union A , show

$$\sum_{n=0}^{\infty} E(X; A_n) = E(X; A),$$

i.e. the sum converges absolutely and has the displayed value.

Knowledge used. [durrett_ch1_expectation_as_integral](#); [durrett_ch1_expectation_linearity](#); [durrett_ch1_expectation_limit_theorems](#).

Solution. Since X is integrable,

$$\sum_{n=0}^{\infty} E(|X|; A_n) = E\left(|X|; \bigcup_{n=0}^{\infty} A_n\right) = E(|X|; A) \leq E|X| < \infty,$$

where we used Exercise 1.6.15 for the nonnegative random variables $|X|\mathbf{1}_{A_n}$. Hence the series $\sum_n E(X; A_n)$ converges absolutely.

Now use finite additivity and pass to the limit:

$$\sum_{n=0}^N E(X; A_n) = E\left(X; \bigcup_{n=0}^N A_n\right).$$

As $N \rightarrow \infty$,

$$X\mathbf{1}_{\bigcup_{n=0}^N A_n} \rightarrow X\mathbf{1}_A \quad \text{and} \quad \left|X\mathbf{1}_{\bigcup_{n=0}^N A_n}\right| \leq |X|.$$

Dominated convergence gives

$$E\left(X; \bigcup_{n=0}^N A_n\right) \rightarrow E(X; A).$$

Therefore

$$\sum_{n=0}^{\infty} E(X; A_n) = E(X; A).$$

Exercise 1.7.1

Question. Show that if

$$\int_X \int_Y |f(x, y)| \mu_2(dy) \mu_1(dx) < \infty,$$

then

$$\int_X \int_Y f(x, y) \mu_2(dy) \mu_1(dx) = \int_{X \times Y} f d(\mu_1 \times \mu_2) = \int_Y \int_X f(x, y) \mu_1(dx) \mu_2(dy).$$

Knowledge used. [durrett_ch1_product_measure](#); [durrett_ch1_fubini_tonelli](#); [durrett_ch1_integrable_function](#).

Solution. The hypothesis says that one iterated integral of $|f|$ is finite. By Tonelli's theorem applied to the nonnegative function $|f|$,

$$\int_{X \times Y} |f| d(\mu_1 \times \mu_2) = \int_X \int_Y |f(x, y)| \mu_2(dy) \mu_1(dx) < \infty.$$

Thus f is integrable with respect to the product measure. Fubini's theorem applies to f , and gives both iterated integrals and their equality to the product integral:

$$\int_X \int_Y f d\mu_2 d\mu_1 = \int_{X \times Y} f d(\mu_1 \times \mu_2) = \int_Y \int_X f d\mu_1 d\mu_2.$$

The stated corollary follows by taking $X = \{1, 2, \dots\}$ with counting measure and considering $f(n, \omega) = f_n(\omega)$. The assumption $\sum_n \int |f_n| d\mu < \infty$ is exactly the integrability hypothesis above, so

$$\sum_n \int f_n d\mu = \int \sum_n f_n d\mu.$$

Exercise 1.7.2

Question. Let $g \geq 0$ be measurable on $(X, \mathcal{A}, \mu)$. Use Fubini/Tonelli to show

$$\int_X g d\mu = (\mu \times \lambda)(\{(x, y) : 0 \leq y < g(x)\}) = \int_0^\infty \mu(\{x : g(x) > y\}) dy.$$

Knowledge used. [durrett_ch1_fubini_tonelli](#); [durrett_ch1_sections_measurable](#); [durrett_ch1_tail_sum_nonnegative](#).

Solution. Let

$$E = \{(x, y) : 0 \leq y < g(x)\}.$$

Since g is measurable, E is measurable in the product σ -field. For each fixed x ,

$$\lambda(E_x) = \lambda(\{y : 0 \leq y < g(x)\}) = g(x).$$

Therefore Tonelli's theorem gives

$$(\mu \times \lambda)(E) = \int_X \lambda(E_x) \mu(dx) = \int_X g(x) \mu(dx).$$

For each fixed $y \geq 0$, the y -section is

$$E^y = \{x : g(x) > y\}.$$

Thus another application of Tonelli gives

$$(\mu \times \lambda)(E) = \int_0^\infty \mu(\{x : g(x) > y\}) dy.$$

Combining the two identities proves the formula.

Exercise 1.7.3

Question. Let F, G be Stieltjes measure functions and let μ, ν be the corresponding measures. Show:

(i)

$$\int_{(a,b]} (F(y) - F(a)) dG(y) = (\mu \times \nu)(\{(x, y) : a < x \leq y \leq b\}).$$

(ii)

$$\int_{(a,b]} F(y) dG(y) + \int_{(a,b]} G(y) dF(y) = F(b)G(b) - F(a)G(a) + \sum_{x \in (a,b]} \mu(\{x\})\nu(\{x\}).$$

(iii) If $F = G$ is continuous, then

$$\int_{(a,b]} 2F(y) dF(y) = F^2(b) - F^2(a).$$

Knowledge used. [durrett_ch1_stieltjes_measure](#); [durrett_ch1_product_measure](#); [durrett_ch1_fubini_tonelli](#).

Solution.

(i) For $y \in (a, b]$,

$$F(y) - F(a) = \mu((a, y]).$$

Hence, by Tonelli,

$$\begin{aligned} \int_{(a,b]} (F(y) - F(a)) dG(y) &= \int_{(a,b]} \int_{\mathbb{R}} \mathbf{1}_{\{a < x \leq y\}} \mu(dx) \nu(dy) \\ &= (\mu \times \nu)(\{(x, y) : a < x \leq y \leq b\}). \end{aligned}$$

(ii) Let

$$D = \{(x, y) : a < x \leq y \leq b\}.$$

Part (i) gives

$$(\mu \times \nu)(D) = \int_{(a,b]} F(y) dG(y) - F(a)(G(b) - G(a)).$$

Similarly,

$$(\mu \times \nu)(D') = \int_{(a,b]} G(x) dF(x) - G(a)(F(b) - F(a)),$$

where

$$D' = \{(x, y) : a < y \leq x \leq b\}.$$

The two sets D and D' cover the rectangle $(a, b] \times (a, b]$ and overlap exactly on the diagonal $\{(x, x) : x \in (a, b]\}$. Therefore

$$(\mu \times \nu)(D) + (\mu \times \nu)(D') = (F(b) - F(a))(G(b) - G(a)) + \sum_{x \in (a,b]} \mu(\{x\})\nu(\{x\}).$$

Substituting the two integral expressions and simplifying gives the claimed identity.

(iii) If $F = G$ is continuous, then $\mu(\{x\}) = 0$ for every x , so the jump sum in part (ii) is zero. Setting $G = F$ gives

$$2 \int_{(a,b]} F(y) dF(y) = F^2(b) - F^2(a).$$

Exercise 1.7.4

Question. Let μ be a finite measure on $\mathbb{R}$ and

$$F(x) = \mu((-\infty, x]).$$

Show that, for $c > 0$,

$$\int_{\mathbb{R}} (F(x+c) - F(x)) dx = c \mu(\mathbb{R}).$$

Knowledge used. [durrett_ch1_stieltjes_measure](#); [durrett_ch1_fubini_tonelli](#); [durrett_ch1_integrable_function](#).

Solution. For each x ,

$$F(x+c) - F(x) = \mu((x, x+c]).$$

Therefore

$$F(x+c) - F(x) = \int_{\mathbb{R}} \mathbf{1}_{(x, x+c]}(t) \mu(dt).$$

By Tonelli's theorem,

$$\begin{aligned} \int_{\mathbb{R}} (F(x+c) - F(x)) dx &= \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_{(x, x+c]}(t) \mu(dt) dx \\ &= \int_{\mathbb{R}} \lambda(\{x : x < t \leq x+c\}) \mu(dt). \end{aligned}$$

For fixed t , the set $\{x : x < t \leq x+c\}$ is $[t-c, t)$, which has Lebesgue measure c . Hence

$$\int_{\mathbb{R}} (F(x+c) - F(x)) dx = \int_{\mathbb{R}} c \mu(dt) = c \mu(\mathbb{R}).$$

Exercise 1.7.5

Question. Show that $e^{-xy} \sin x$ is integrable on the strip $0 < x < a$, $0 < y < \infty$. Evaluate the double integral in both orders and obtain the usual sine-integral estimate.

Knowledge used. [durrett_ch1_fubini_tonelli](#); [durrett_ch1_integrable_function](#); [durrett_ch1_dominated_convergence](#).

Solution. First,

$$\int_0^a \int_0^\infty e^{-xy} |\sin x| dy dx = \int_0^a \frac{|\sin x|}{x} dx < \infty,$$

because $|\sin x|/x$ is bounded near 0 and integrable on $[0, a]$. Thus Fubini's theorem applies.

Integrating first in y gives

$$\int_0^a \int_0^\infty e^{-xy} \sin x dy dx = \int_0^a \frac{\sin x}{x} dx.$$

Integrating first in x , we use

$$\int_0^a e^{-xy} \sin x dx = \frac{1 - e^{-ay}(y \sin a + \cos a)}{1 + y^2}.$$

Therefore

$$\int_0^a \frac{\sin x}{x} dx = \int_0^\infty \frac{dy}{1+y^2} - (\cos a) \int_0^\infty \frac{e^{-ay}}{1+y^2} dy - (\sin a) \int_0^\infty \frac{ye^{-ay}}{1+y^2} dy.$$

Since $\int_0^\infty (1+y^2)^{-1} dy = \pi/2$, this is

$$\int_0^a \frac{\sin x}{x} dx = \frac{\pi}{2} - (\cos a) \int_0^\infty \frac{e^{-ay}}{1+y^2} dy - (\sin a) \int_0^\infty \frac{ye^{-ay}}{1+y^2} dy.$$

Consequently, for $a \geq 1$,

$$\begin{aligned} \left| \int_0^a \frac{\sin x}{x} dx - \frac{\pi}{2} \right| &\leq \int_0^\infty e^{-ay} dy + \int_0^\infty ye^{-ay} dy \\ &= \frac{1}{a} + \frac{1}{a^2} \leq \frac{2}{a}. \end{aligned}$$

Note. The extracted exercise text displayed $\arctan(a)$ in the place where the computation over the stated strip gives $\pi/2$. With the stated strip $0 < x < a$, $0 < y < \infty$, the formula above is the standard Fubini computation. Since $|\pi/2 - \arctan(a)| \leq 1/a$, it also gives the related bound

$$\left| \int_0^a \frac{\sin x}{x} dx - \arctan(a) \right| \leq \frac{3}{a} \quad (a \geq 1).$$